

Industrial Laser Cutter

Operation Checklist

1. Add your name, date, time and material to be cut in the operation log.
2. Turn on water chiller
3. Obtain key from lockbox with authorized code and use key to unlock machine
4. Turn on power to controller (top switch) on right side of machine. The controller will auto-home the laser head.
5. Place work piece inside cabinet and adjust Z axis to ensure it is below the head.
6. Set Z-distance from laser carefully using the green Z up/down buttons. This is normally a 5mm gap between the laser head and work piece although depends on the focal length of the lens. Use the step gauge block to set gap distance.
7. Ensure work bed is level, especially if using honeycomb bed to support work piece. If not using honeycomb bed, ensure slats are level and adequately support cut material.
8. Make sure you know the location of the CO2 extinguisher and fire blanket (on top of water chiller)
9. CLOSE ALL PANELS to avoid laser exposure.
10. Load job onto machine either through PC or USB stick. Job should never specify a laser power of over 50% because it significantly reduces the life of the laser tube. When cutting jobs, it should be for a RUIDA controller G-code.
11. Open chimney flu gate located on exhaust tube above back of machine.
12. Turn on power to exhaust fans (located near the breaker panel).
13. Verify water temperature is between 16 and 25 degrees C. Top off with distilled water if level is low.
14. Turn on laser power supply (middle switch on right side of machine).
15. Turn on air assist (pump/compressor).
16. Select file with ENTER button and press FRAME button to preview cutting area
17. Press green "Start/Pause" button to start cutting.
18. When job is finished, remove work piece. Controller will beep when job is finished.
19. In case of issues or incidents during cutting, please place appropriate sign on machine and log failure with job log sheet found in manila folder by machine.
20. Turn off air assist.
21. Close chimney flu gate (above machine)
22. Turn off exhaust fans if no other vented equipment is in use.
23. Turn off laser tube and controller power.
24. Turn off water chiller.
25. Leave the machine clean: empty chip tray and wipe down residue from cut.
26. Return lock-out key to lock-box.